Course Essentials

Summary

- CS-290, Cécile Hardebolle. **Flipped class** : theory at home (videos + quizzes, about 5 videos/week), practice in class.
- **Tuesday 10 :15-12 :00** : programming exercise (Jupyter notebook on Noto).
- **Monday 15 :15-17 :00** : review questions (ttpoll) + case studies.
- Solutions are provided for notebooks and case studies.

Contents of the course

- **Safety** : harmful impacts, safety & security, micro/meso/macro.
- **Fairness** : sources and impacts of biases, fairness measures.
- **Sustainability** : environmental impact, life cycle assessment.
- **Empowerment** : user agency, transparency, explainability.
- **Privacy** : data traces, data protection, re-identification, surveillance.
- **Strategies** : stakeholder analysis, impact anticipation, values elicitation, risk assessment, decision-making.

Explained to a goat

You watch the videos at home and you practise in class, with help. The course is about the ways software hurts people, and about methods to see it coming.

Evaluation

Key idea

- **Mini-project 5%** : groups of 2, 4 deliverables (2-3 pages), no programming, one in four graded. **Questions on it at the final exam.**
- **Mid-term 25%** : 2 November, 15 :15, 1h30, pen and paper, individual, **one A4 sheet** (both sides, free format). Chapters Introduction, Safety 1-2, Fairness 1-2. Single-choice + short open questions; notebooks (code reading, output, reflection) and theory.
- **Final 70%** : winter session, same rules, whole course.

Warning

GenAI is prohibited for all graded work. Answers can be in French. Arguments must be understandable : 1 point ≈ 1-2 complete sentences.

Learning loop : **Experience** (problem to solve with programming) + **Knowledge** (theory + strategies) + **Habits** (strategies applied to cases).

Explained to a goat

Three grades : a tiny project, a mid-term and a big final, always with one cheat sheet and no ChatGPT.

AI Incidents and Ethical Debt

Example

OECD AI Incidents and Hazards Monitor (AIM)

- Collects AI incidents (harm happened) and hazards (harm could happen) reported in the press.
- The **number** of incidents and hazards is **increasing**; their **proportion** among AI events is rather stable.
- Limits : depends on what the press reports (countries, languages, spectacular cases).

Definition

ETHICAL DEBT (Petrozzino, 2021)
The **price paid by people and the environment** because of systems **released with ethical flaws**.

Key idea

Causes of ethical debt : engineers care about ethics, but

- competition, time pressure, organisational incentives;
- inherited practices and culture, decision-making power;
- beliefs about tool neutrality, lack of ethical knowledge;
- not seeing the ethical implications of technical decisions = **"veiled" ethical agency** (Herkert 2020, Widder 2023).

This is what the course is about.

Explained to a goat

Ship fast with ethical holes, and someone else pays later. It happens mostly because engineers do not *see* that their technical choices are also ethical choices.

Weekly Summary

Summary

- Flipped class : videos at home, notebooks on Tuesday, cases on Monday.
- Grades : project 5%, mid-term 25% (2 Nov), final 70%; one A4 sheet.
- **Ethical debt** = price paid by people and environment for systems released with ethical flaws.
- Main cause : **veiled ethical agency**.